

How to Write a Visualization Research Paper: A Starting Point

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Abstract

This paper attempts to explain the mechanics of writing a research paper in visualization. This serves as a useful starting point for those who have never written a research paper before or have very little previous experience. After all, no one is born knowing how to write one. And yet, there are certain elements, a commonality, that should be found in virtually all good visualization research papers. We give our recommendations as to each section a good research paper consists of as well as what each section contains. This paper itself follows our recommended structure.

We believe that paper writing should start with the abstract. The abstract can be approximately 6–12 sentences. It is a difficult starting point, but it forces the author to write down a concise description of what they are researching and what the benefits are. Chances are, if the author cannot start out by writing an abstract, then it is not clear in the author's mind what the paper should be about. Of course, the abstract will be refined and updated during the paper writing process. The abstract should concisely (1) identify the research topic, (2) describe the novelty of the presented work and (3) identify the benefits and advantages that result.

Keywords: visualization research paper, writing methodology

ACM CCS: K.3.2 [Computer and Information Science Education]: Computer Science Education

1. Introduction (Motivation)

'Give a man a fish; you have fed him for today. Teach a man to fish; and you have fed him for a lifetime.'—Lao Tzu (Chinese proverb).

This quote summarizes the idea behind this paper. It strives to serve as a fishing rod, or at least partially, for those new to writing research papers. These brief guidelines are meant to serve as a *starting point* to those (most likely students) writing a research paper for the first time. The idea was inspired by some new PhD candidate students studying visual computing and preparing their first research paper(s). Writing a good visualization research paper is difficult, yet, students are given this task with little or no instruction. We felt a strong need to at least provide the student with some written guidelines to start with, which, to the best of our knowledge, do not exist.

Here, we go through section-by-section, those components that we believe a good research paper, in general, contains. We give general informal guidelines on the contents of each section as well. We describe the *Introduction* (Section 1), *Related Work* (Section 2), *Method* (Section 3), *Implementation* (Section 5), *Results* (Section 6), *Conclusions and Future Work* (Section 8) and *Acknowledgements* (Section 9), sections. We also provide recommendations on paper titles (Section 1), temporal planning (Section 1), literature searching (Section 2), collaborations (Section 2), diagrams and figures, images (Section 3.5), enhancements (Section 4), application papers (Section 7), performance analysis (Section 6.2), movies (Section 6.3), supplementary material (Section 6.3), proofreading (Section 9.1) and references (Section 9.2). Several citations to examples are provided for reference. We keep in mind that these topics are rather subjective and our discussion of each topic is driven by experience. The result

is only meant to serve as a rough guideline and a useful starting point. There is no substitute for experience or an experienced advisor. This paper itself is structured according to our recommended guidelines.

Although the title of the starting section is 'Introduction' it should really be *Motivation*. Quickly, in one or two sentences, introduce the topic. This is followed immediately with *why* this is a useful research direction, including possible applications of the presented research. 'What is this research topic good for?' is really the question a good introduction attempts to answer. See Chen *et al.* [CLZ07] for a good example of motivating the presented research.

After the motivation behind the subject is described, the novelty, benefits and advantages of the research are described. Remember, this is actually where a research project often starts. It begins with an idea. And one key to evaluating how good an idea is, is being able to identify its novelty and advantages. Each contribution and each benefit that reading this paper provides is identified very explicitly in a bulleted list. Possible points to mention include

- Does the presented algorithm(s) provide a novel visualization of some sort? That is, images never before seen.
- Does the presented work provide new interaction techniques not previously published?
- Does the technique described result in faster performance times than previous algorithms and if so why?
- Does the approach provide new insight in some way?
- Does the work take advantage of the graphics card hardware in some way? For example, using texture-mapping hardware, or hardware-accelerated interpolation?
- What are the contributions, precisely, of this paper? What does the reader potentially gain from this paper?

Words such as *contribute*, *novel* and *advantageous* are used in this list. An explicit bulleted list should be used to make the subject black and white to the reviewer, otherwise, a reviewer might not understand the contribution described. See Chen *et al.* [CMLZ08], Garth *et al.* [GLT*07] and Zhang *et al.* [ZYZL09] for good examples of such lists of contributions.

The presented work normally builds on a previous algorithm. If so, your work should inherit benefits from the previous work. Those inherited advantages may also be listed. See Grabner and Laramée [GL05] for a good example of a bulleted list of inherited benefits.

To achieve the amazing benefits your research results have to offer, specific challenges must be overcome, both conceptual and technical. Those challenges can be mentioned here very briefly.

The 'Introduction' section then concludes with how the rest of the paper is organized. Section 2 presents previous

work on this topic. Section 3 presents the method starting with an overview. Section 5 describes some unexpected aspects of the our model necessary for a successful implementation. Section 6 presents our results both visual and from a performance point of view. Section 7 addresses application papers. Section 8 completes the exposition with conclusions and future work.

1.1. On paper titles

Regardless of the title of your paper, in general it will informally be referred to using two to three words. For example, 'Marching Cubes: A High Resolution 3D Surface Construction Algorithm' is simplified to 'The Marching Cubes' paper [LC87]. 'Visual Analysis and Exploration of Fluid Flow in a Cooling Jacket' becomes 'The Cooling Jacket' paper [LGD*05]. On one hand, the title of your paper should ideally be simple enough such that it can easily be remembered and identified with two or three words. On the other hand, a title should also be long enough to convey some sort of helpful meaning and description to the reader. This problem can usually be resolved using a *title:subtitle* combination, as in the case of Marching Cubes [LC87] (no pun intended).

1.2. On temporal planning

We believe a high visibility publication requires approximately 2 months to write. A good, complete first draft takes about 1 month to write. During the second month, the paper can be improved with approximately four rounds (or more) of iterative feedback from a co-author(s). Taking this point of view, paper writing should start about 2 months before the proposed deadline. Extra time can be allowed for non-native speakers.

1.3. On LaTeX

In our opinion, the best software for writing papers is $\text{\LaTeX}$. $\text{\LaTeX}$ produces high quality, professional-looking output. Virtually all conferences and journals provide $\text{\LaTeX}$ templates for paper submissions. We encourage unfamiliar readers to explore this software. Kopka and Daly provide an excellent $\text{\LaTeX}$ reference guide [KD04].

2. Related Work

The 'Related Work' section is actually what makes a research paper a research paper. It demonstrates to the reader that you have done your homework, reviewed the previous literature, and now are ready to present a new contribution over what has been previously published.

One of the difficult aspects of the Related Work section is choosing the proper scope, for example, how many related papers to include and which ones. For new authors, this is

something an advisor can help with. In short, each previous publication gets a two to three sentence description. In the description it is made clear to the reader how the presented work (this paper) is different (better) than the previous piece of literature.

It is important to note that there is some subjectivity in choosing which papers to refer to and also importantly, which previous literature not to refer to. In particular, the references contained in the paper may influence who is chosen to review the paper. Some senior reviewers, papers chairs, or editors may choose reviewers based on the authors appearing in the list of references (a logical and common method). This means that you as a writer may also influence (but by all means not deterministically) who ends up reviewing your paper. Given a complete list of references, a sub-set of them can be considered mandatory and some can be considered optional. Thus, some authors use the optional references to influence who will review the paper and perhaps will also try to steer away from certain reviewers.

When referring to previous work use names, not only numbers (or initial letters). So, instead of writing, 'In [JL97], an algorithm for the creating of evenly-spaced streamlines is presented', write, 'Jobard and Lefer present an algorithm for creating evenly-spaced streamlines [JL97]'. The names are much more helpful to the reader than the numbers alone. Also, the previous literature was written by real people whom, in the best case scenario, you will actually meet at a conference. Writing the names also shows a certain respect and appreciation for the previous work that numbers alone do not.

Incidentally, this paper also has related literature. Smith describes how a manuscript should be refereed [Smi90]. Lee presents another interesting paper on how to review a research paper [Lee95]. Cetintemel has written an excellent article about how to conduct an academic job search targeted at computer scientists [Cet01]. Globus and Raible have written a great how-to-cheat when writing a visualization paper [GR94]. Munzner [Mun08] provides a nice overview of ways to prevent an information visualization paper from being rejected from the review process. We note this paper is targeted towards scientific visualization papers, however, we believe it is also a helpful starting point for those preparing information visualization manuscripts.

2.1. On the literature search

For a visualization paper, the basic previous literature search consists of looking through the previous 10 years of (1) the IEEE Visualization conference proceedings, (2) the IEEE Transactions on Visualization and Computer Graphics (IEEE TVCG) journal papers and (3) the EuroVis conference proceedings (called VisSym until 2005). In principle it is simple, but it is time consuming. From these three publication venues basically all other related work can be found by look-

ing through the references of the related papers found in the three aforementioned publications. First, prospective related papers are found by reading the titles. After candidates are found based on titles, then their abstracts should be read. After candidates are found based on abstract, then the full papers should be browsed.

It is also important to look for survey papers and state-of-the-art reports. Very helpful state-of-the-art reports are published every year at the EUROGRAPHICS conference and also in the *Computer Graphics Forum* journal. There is also a chance that a related survey paper has been published in the *ACM Computing Surveys* journal.

The literature search is an educational process. If you find hundreds of previous papers that appear to be related to the chosen topic, this may be a warning sign. For example, if you chose to work on isosurface rendering, then you would, in fact, find hundreds of previous papers on this topic. That means that this is already very well researched. Thus, the chances of making a contribution in this area with impact are reduced. (We learned this the hard way [LB02].) It is better to find a topic with a more manageable, preferably small, amount of previous literature. How to read a research paper is another topic [Lar10].

2.2. On collaborations

Collaborators are good for filling in the gaps when you suspect that there may be holes or an absence of knowledge somewhere. We think that collaborations, in general, have a tendency to increase the quality of a paper. Also, we find collaborations one of the best ways of learning new things and for networking.

3. Method (or Computational Model)

Section 3 is the main content of the paper and thus occupies the most space. It starts with an overview (Section 3.1) followed by the details of each component of your system or each stage of your method, for example Section 3.2 and so on. The name of Section 3 could be the name of your algorithm (if it has a name, e.g. Image Space Advection (ISA) [LvWJH04]). Section 3 could also be called just *Method* or *Computational Model*. It presents an idealized model or process by which your algorithm or system works (or how it should work).

3.1. Method overview

The description of any system or algorithm should start out with an overview before jumping into the details. An overview should include an overview diagram like that of Figure 1. The method overview diagram depicts the *conceptual model* that the algorithm or system in this paper follows.

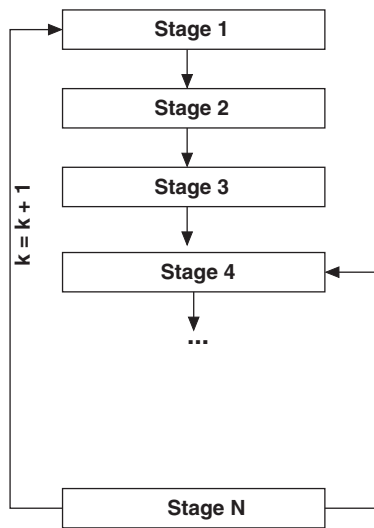


Figure 1: *Method overview.* An overview diagram illustrates the conceptual model your algorithm implements. Figures should be placed at the top of the page by default. This figure was made using Xfig. Inkscape is another good, open source tool for producing diagrams.

This is not to be confused with its implementation. Of course there is a correspondence, however, a single concept has potentially many different implementations. This subsection focuses on the concept. Other subsections that follow such as Sections 3.2 and 3.3 can describe implementation details, including pseudocode. See Chen *et al.* [CMLZ08], Laramée [Lar03] and Schultz and Seidel [SS08], for examples of pseudocode describing the method. One nice property of Schultz and Seidel's pseudocode is that it fully describes the input and output of the algorithm.

Section 3, like paper writing, can follow a divide-and-conquer approach. Just as a paper is divided up into sections, Section 3 can be broken down into subsections—starting with an overview diagram. Each subsection in Section 3 can then correspond to a component of the overview diagram.

See Gyulassy *et al.* [GBHP08], Jobard *et al.* [JEH01], Laramée *et al.* [LvWJH04], Spencer *et al.* [SLCZ09] and Van Wijk [vW05] for good examples of overview diagrams. Gyulassy *et al.* [GBHP08] provide a really nice diagram that exploits computer graphics to provide a concise step-by-step algorithm guide. See Laramée *et al.* [LHD*04] for overview diagrams used in a state-of-the-art literature survey.

3.2. Stage 1

For each stage or component of your system or algorithm there should be a corresponding subsection in Section 3 providing further details to the reader. Subsections themselves

may contain further diagrams, illustrations, examples and pseudo-code.

3.3. Stage 2

A description of the next component in the presented algorithm or system (from Figure 1) is presented here.

3.4. Parameters

Any new algorithm ultimately introduces new parameters, for example, threshold values, α values, special distances, minimum and maximum values, etc. Identifying and discussing these new parameters is a good idea. Illustrating the effect of setting these parameters to a range of different values is instructive and helps the reader gain an understanding of the algorithm or method being presented. In fact, during any implementation, the best value of any new parameter is generally unknown. Thus, it is best to implement them as user options since their value may change depending on the data set being evaluated. After a thorough testing phase, the new parameter may be given an optimal default value. See Jobard *et al.* [JL97, JEH01, JEH02], Van Wijk [vW02] and Telea and Van Wijk [TvW99], for good example discussions of new algorithm parameters.

3.5. Figures and images

Figures, images and their captions should be 'self-contained'. By self-contained, we mean that the reader should be able to look at any figure or image, read its caption, and gain a basic understanding *without* having to read the main text. Enough details should be provided in the caption such that the reader is not required to read the whole paper to understand an image or figure. Many reviewers (and readers), will make a first pass through a paper simply by flipping through the pages, looking at the images and captions, and already gain an impression of the paper. This first pass may already influence the final reviewer's opinion and thus the outcome of the review. If the reviewer appreciates the images and can gain an understanding of them from the captions, they may form a favourable first impression of the paper. This is true about readers in general and does not only apply to research papers. When a potential reader picks up any reading material, whether it be a newspaper, magazine or book, they first flip through it to look at the pictures. If they find an interesting picture, they read the caption. If they find interesting pictures and captions, they may then go on to read the main text.

Figures and images that show new visualizations are very good to include in a visualization paper. In fact, that is a major goal. If an image really shows something new, then it's probably a good figure to include. Pedagogic figures are also good. On the topic of figure quality, probably only

experience from reading other papers can help in judging the proper level of quality.

The author will make his job easier if all of his images, for example screen shots or frame buffer images, are the same resolution, perhaps through a user setting. Also, a user setting that changes the background colour to white is very useful. Images with a white background are best for papers in our opinion. We refer the reader to Tufte [Tuf83] for further stylistic guidelines.

4. Enhancements (or Extentions)

After the main details of your research method are presented in Section 3, a follow-up section may be added that includes enhancements or extentions to the basic algorithm. There may be a subsection for each enhancement or extention. See Jobard *et al.* [JEH02], Jobard and Lefer [JL97] and Turk and Banks [TB96] for examples of this. Enhancements or extentions do not have to be a separate section of the paper. They could be subsections of Section 3.

5. Implementation

This section is more-or-less optional. It describes the implementation of your model from Section 3. In particular, it describes aspects of your model that are *unexpected*. In other words, during the implementation of the computational model, what things came up that were unexpected. What had to be implemented in order for the model to work? Was the implementation more difficult than expected? Or is it easy? Sometimes the implementation details are also integrated into the corresponding method subsections.

6. Results and Performance

The ‘Results’ and ‘Performance Analysis’ sections can be separate, but they are often lumped together in the same section. If the performance analysis is more extensive it can be put in its own section. Separate subsections are also logical.

6.1. Results

In this section, the nicest images of your algorithm are presented to both synthetic and real-world data sets. The (new) insight provided by the visualization is described. A concise description of the data set the algorithm is being applied to is also described. The following characteristics of the data sets should be given: (1) what the data sets are and perhaps what domain they come from, (2) how large the data sets are—or their resolution in the case of structured data, (3) the spatial and temporal dimensionality of the data sets should be given, (4) whether or not they are represented by structured or unstructured grids and (5) if they are multi-resolution or adaptive resolution data samples.

Hopefully, you will be able to show your algorithm being applied to data sets from another domain, for example, the visualization of simulation data, to demonstrate that your algorithm actually works in general.

6.2. Performance analysis

In visualization (and computer graphics), there’s always a trade-off between quality (or accuracy) and speed (or memory and speed). Generally, the faster the performance, the lower the quality. The algorithm or system being presented should have some free parameters that can be adjusted to trade-off quality for speed and vice-versa. Essentially, *every* parameter whose value you are not sure of should be a user-defined parameter—an option that can actually be changed by the user at run time (not just compile time).

Remember, visualization is used for analysis, exploration and presentation. For presentation, the image quality must be at its highest, and thus performance probably at its slowest. The highest quality is used to generate the images for the paper, presentation and videos. For exploration, the algorithms and systems are tuned for the fast performance times possible. All optional enhancements should be turned off (unless your algorithm always runs at interactive frame rates).

A table of performance times can report both aspects, optimized for speed, that is exploration mode, and optimized for quality, that is presentation mode. The best and most honest reports of performance times will show when the algorithm or system reaches its limits, that is when it starts to run slow due perhaps to large data sets. Algorithms also usually trade-off memory for speed. In other words, faster algorithms often require more memory, basically because more calculations are pre-processed and stored. Unfortunately, these limits are often left out of visualization research papers.

See Laramée *et al.* [LB02, LvWJH04], Peng and Laramée [PL08] and Spencer *et al.* [SLCZ09] for examples of detailed performance analysis that trade- off between performance and quality or memory and speed.

6.3. Movies and supplementary material

The submission of a research (or application) paper is generally strengthened by an accompanying MPEG movie or animation. Movies are especially necessary to communicate time-dependent aspects of an algorithm or system that cannot be conveyed in static images alone. This includes aspects such as interaction, animation in a visualization and general time-dependent phenomena. An accompanying movie may also give the reviewer an impression that the work is polished (as opposed to something that was thrown together an hour before the deadline). Movies can also be exploited to show results that did not fit into the paper due to size restrictions. It is also helpful if a supplementary movie demonstrates some

of the parameters discussed in the paper (as discussed in Section 3.4).

We strongly recommend saving and submitting a supplementary movie in MPEG format, even if it means less compression. At the time of this writing, MPEG is the only platform-independent movie format available. And this has been the case for the last ten or so years. Regardless of what kind of computing environment the researcher is in, the readers and reviewers come from a *wide* variety of backgrounds spanning multiple generations (both in human and computer life spans) and will generally be using a wide variety of computing platforms. Some reviewers are likely to still be using old SGI machines (for example) simply for sentimental reasons. If they cannot view your supplementary movie at the time of review, the opportunity is lost.

In terms of the mechanics, we recommend adding a user option to your software or system that automatically: (1) re-sizes the viewer to 512² pixels (Old MPEG players can only handle movie resolutions which are a power of two.) and (2) saves each frame as a still image in JPEG (or PNG) format. The still images are used as input to an application which converts them to MPEG format. We use *Adobe Premiere* (<http://www.adobe.com/products/premiere/>) because of its rich set of features. *Adobe Premiere* is *expensive* so there are free, alternatives such as *VideoMach* (<http://www.gromada.com/videomach.html>). However, *VideoMach* is not so feature-rich. The still images can then be edited, annotated and re-ordered (etc.), as needed. The MPEG video can either be uploaded using the same submission system provided by the conference or journal. Alternatively, a URL may be provided in the paper itself pointing to the video in the case that no submission option is available. Don't forget to check the maximum submission file size when compiling a video. If the maximum file size submission seems to low, a request can always be made to increase it.

See the author's web site for several sample supplementary videos.

7. Application Papers

An application paper presents the contribution of visualization techniques towards the understanding of application-specific data. This class of paper examines and discusses the effectiveness of the visualization methods for a particular application. In other words, an application paper describes how known, off-the-shelf visualization techniques have been used (and possibly adapted) to study a domain-specific problem. It is not essential to present original visualization techniques. Although there is overlap, in terms of content and layout, with research papers, there are also differences.

Section 1, the *Introduction*, is very similar for an application paper. It provides the motivation for studying the topic at hand.

Section 2, *Related Work*, is also very similar to a normal research paper. However, in the case of an application paper, there are also additional related research papers from the application domain being studied. For example, if the application paper studies a problem from the biological sciences, then the related work section may also contain references from the biological sciences that may not normally be featured in a visualization research paper.

An application paper may contain an additional section 3, *Background*. The Background section may describe some pre-requisite knowledge stemming from the application domain being studied. In other words, the Background section may provide information not presumed to be known *a priori* to a typical computer scientist, but may be well-known to a scientist from the application domain. One of the important roles of the Background section is to describe explicitly the questions that the domain-specific scientist is trying to answer when carrying out their work. See Laramée *et al.* [LSH04, LWSH04, LGD*05] for examples of these types of questions.

Instead of the *Method* section found in a research paper, an application paper contains a *Visualizations* section. The Visualizations section describes and presents all of the visualizations that were applied to the domain-specific problem. It also details the additional advantages or insights that each visualization has to offer over previous studies of the same problem. Often, other disciplines will use simple visualizations to study their data such as time-series plots, 2D scatter plots, and bar charts, etc. The Visualizations section illustrates the advantages of your techniques over the traditional, simple visualizations. The Method section should also carefully describe how a particular visualization technique has been adapted to be suitable to the problem at hand. See Scheuermann *et al.* [SBH99] for an example of this.

Instead of the *Results* section normally found in a research paper with a traditional performance analysis, an application paper presents a *Domain Expert Review* section. See Grundy *et al.* [GJL*09] for an example of domain expert review. The two most common criticisms of an application paper are that (1) no additional insight is provided by the visualizations used by the author (over those used previously) and (2) no feedback has been given by an expert from the application domain. Domain expert feedback is an essential ingredient in an application paper. The visualizations must be shown to a domain expert and the domain expert should provide some feedback.

The remaining Conclusions and Future Work sections are similar to a research paper.

8. Conclusions and Future Work

The Conclusions section simply summarizes both the novelty of the presented work and lists the resulting advantages. Remind the reader why the presented work is so

good and what's new. Basically, it's like a concise, one-to-two paragraph summary of the abstract and introduction sections.

The future work should be short. Generally it's about three sentences of possible future extensions to the presented work. If you're not sure what to write, it's almost always possible to include a GPU-centred version of the presented work. Extending the visualization to a higher dimensionality is also often a future work project.

Example: We have presented the essentials to writing a research paper in visualization. Although every research paper is different, there is some commonality that can be found in all good research papers. Here we attempt to identify that commonality and describe the essential necessary ingredients of paper sections and their contents. Several example research papers are cited for reference. The result is meant to serve as a starting point for those with little or no experience in writing research papers. Writing research papers is both an art and an engineering exercise and no one is born knowing how.

Future work includes writing an HCI version of this paper and writing a more general version that applies to computer science research papers.

9. Acknowledgements

The Acknowledgements section thanks any agency or organization that may have contributed funding to the presented research, for example the EPSRC, KPlus, the NIH, the NSF, etc. This section also mentions any people that may have contributed in some way, for example with the implementation or with discussions or feedback.

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There is one other thing the Acknowledgements section can be used for and that is to thank potential reviewers. If you have a strong suspicion about who will end up reviewing your paper, then it does not hurt to put this person's name in the Acknowledgements section and thank them in some capacity. See Laramée *et al.* [LSH04] for an example of this used in practice. The first author had a strong suspicion that Van Wijk would be a reviewer of that paper and acknowledged his contributions before the review process.

Questions, comments, suggestions for improvement, contributions to this paper, or any other feedback is not only welcome but encouraged. Please contact the first author.

9.1. On proofreading

In the ideal world, a paper is proofread by someone outside the project the day before submission to look for small typographical errors. This manuscript has gone through this process. Fresh new eyes that do not necessarily know the topic can be good for spotting these things, at least in our experience.

My former advisor, Helwig Hauser, helped me out one day by proofreading a paper I had been working on [LW02]. The version of the manuscript he reviewed had mistakes in the references—comments left behind by accident. My advisor pointed out the errors by saying, 'Bob, . . . you know some people, for example reviewers, *start with the references* when reading a paper, looking them over very carefully'. I thought to myself, 'What kind of anal-retentive geek does that?'

I have turned into one of those people. Whenever I review a paper, I start by detaching the references from the rest of the paper and laying them aside, making it easier to check them as I read through the paper. References should be accurate and complete, that is with page numbers, etc. A paper without complete and correct references can leave a bad impression on the reader and detract from the paper's credibility. Errors can be interpreted as sloppy and also be a sign of hastiness.

9.2. References

- [Cet01] CETINTEMEL U.: On the Academic Interview Circuit: An end-to-end discussion. *SIGMOD Record* 30, 4 (2001), 116–122.
- [CLZ07] CHEN G., LARAMEE R., ZHANG E.: Advanced visualization of engine simulation data using texture synthesis and topological analysis. In *NAFEMS World Congress Conference Proceedings* (May 22–25 2007), NAFEMS, p. 15. (full proceedings on CDROM).
- [CMLZ08] CHEN G., MISCHAIKOW K., LARAMEE R. S., ZHANG E.: Efficient morse decompositions of vector fields. *IEEE Transactions on Visualization and Computer Graphics* 14, 4 (2008), 1–15.
- [GBHP08] GYULASSY A., BREMER P.-T., HAMANN B., PASCUCCI V.: A practical approach to Morse-Smale complex computation: Scalability and generality. *IEEE Transactions on Visualization and Computer Graphics* 14, 6 (2008), 1619–1626.
- [GJL*09] GRUNDY E., JONES M. W., LARAMEE R. S., WISON R. P., SHEPARD E. L. C.: Visualization of sensor data from animal movement. *Computer Graphics Forum*

- (*Proceedings of EuroVis June 10–12, 2009, Berlin, Germany*) 28, 3 (2009), 815–822. (Recipient of the Best Paper Award).
- [GL05] GRABNER M., LARAMEE R. S.: Image space advection on graphics hardware. In *Proceedings of the 21st Spring Conference on Computer Graphics* (May 2005), pp. 75–82.
- [GLT*07] GARTH C., LARAMEE R., TRICOCHÉ X., SCHNEIDER J., HAGEN H.: Extraction and visualization of swirl and tumble motion from engine simulation data. In *Topology-Based Methods in Visualization (Proceedings of Topo-in-Vis 2005)* (2007), Mathematics and Visualization, Springer, pp. 121–135.
- [GR94] GLOBUS A., RAIBLE E.: Fourteen ways to say nothing with scientific visualization. In *Proceedings of the IEEE Computer* (July 1994), pp. 86–88.
- [JEH01] JOBARD B., ERLEBACHER G., HUSSAINI M. Y.: Lagrangian-Eulerian advection for unsteady flow visualization. In *Proceedings of the IEEE Visualization '01* (October 2001), IEEE Computer Society, pp. 53–60.
- [JEH02] JOBARD B., ERLEBACHER G., HUSSAINI Y.: Lagrangian-Eulerian advection of noise and dye textures for unsteady flow visualization. *IEEE Transactions on Visualization and Computer Graphics* 8 3 (2002), 211–222.
- [JL97] JOBARD B., LEFER W.: Creating evenly-spaced streamlines of arbitrary density. In *Proceedings of the Eurographics Workshop on Visualization in Scientific Computing '97* (1997), vol. 7, pp. 45–55.
- [KD04] KOPKA H., DALY P. W.: *A Guide To LaTeX* (4 edition). Addison–Wesley, Boston, MA, 2004.
- [Lar03] LARAMEE R. S.: FIRST: A flexible and interactive resampling tool for CFD simulation data. *Computers & Graphics* 27, 6 (2003), 905–916.
- [Lar10] LARAMEE R. S.: How to read a visualization research paper: Extracting the essentials. In *Proceedings of the IEEE Computer Graphics & Applications (IEEE CG&A)* (2010), forthcoming (available online).
- [LB02] LARAMEE R. S., BERGERON R. D.: An isosurface continuity algorithm for super adaptive resolution data. In *Advances in Modelling, Animation, and Rendering: Computer Graphics International (CGI 2002)* (July 1–5 2002), Computer Graphics Society, Springer, pp. 215–237.
- [LC87] LORENSEN W. E., CLINE H. E.: Marching cubes: a high resolution 3D surface construction algorithm. In *Computer Graphics (Proceedings of ACM SIGGRAPH 87, Anaheim, CA)* (July 27–31 1987), ACM, pp. 163–170.
- [Lee95] LEE A. S.: Reviewing a manuscript for publication. *Journal of Operations Management* 13, 1 (July 1995), 87–92.
- [LGD*05] LARAMEE R. S., GARTH C., DOLEISCH H., SCHNEIDER J., HAUSER H., HAGEN H.: Visual analysis and exploration of fluid flow in a cooling jacket. In *Proceedings IEEE Visualization 2005* (2005), pp. 623–630.
- [LHD*04] LARAMEE R. S., HAUSER H., DOLEISCH H., POST F. H., VROLIJK B., WEISKOPF D.: The state of the art in flow visualization: dense and texture-based techniques. *Computer Graphics Forum* 23, 2 (June 2004), 203–221.
- [LSH04] LARAMEE R. S., SCHNEIDER J., HAUSER H.: Texture-based flow visualization on isosurfaces from computational fluid dynamics. In *Data Visualization, The Joint Eurographics-IEEE TVCG Symposium on Visualization (VisSym '04)* (2004), Eurographics Association, pp. 85–90, 342.
- [LvWJH04] LARAMEE R., VAN WIJK J., JOBARD B., HAUSER H.: ISA and IBFVS: Image space based visualization of flow on surfaces. *IEEE Transactions on Visualization and Computer Graphics* 10, 6 (2004), 637–648.
- [LW02] LARAMEE R. S., WARE C.: Rivalry and interference with a head mounted display. *ACM Transactions on Computer-Human Interaction (TOCHI)* 9, 3 (2002), 238–251.
- [LWSH04] LARAMEE R., WEISKOPF D., SCHNEIDER J., HAUSER H.: Investigating swirl and tumble flow with a comparison of visualization techniques. In *Proceedings of the IEEE Visualization 2004* (2004), pp. 51–58.
- [Mun08] MUNZNER T.: *Information Visualization: Human-Centered Issues and Perspectives, Springer LNCS Volume 4950*. Springer, 2008, ch. Process and Pitfalls in Writing Information Visualization Research Papers, pp. 134–153.
- [PL08] PENG Z., LARAMEE R. S.: Vector glyphs for surfaces: A fast and simple glyph placement algorithm for adaptive resolution meshes. In *Proceedings of Vision, Modeling, and Visualization (VMV) 2008* (2008), pp. 61–70.
- [SBH99] SCHEUERMANN G., BURBACH H., HAGEN H.: Visualizing planar vector fields with normal component using line integral convolution. In *Proceedings of the IEEE Visualization '99* (1999), pp. 255–262.
- [SLCZ09] SPENCER B., LARAMEE R. S., CHEN G., ZHANG E.: Evenly-spaced streamlines for surfaces: An

- image-based approach. *Computer Graphics Forum* 28, 6 (2009), 1618–1631.
- [Smi90] SMITH A. J.: The task of the referee. *Computer* 23, 4 (April 1990), 65–71.
- [SS08] SCHULZ T., SEIDEL H.-P.: Estimating crossing fibers: A tensor decomposition approach. *IEEE Transactions on Visualization and Computer Graphics* 14, 6 (2008), 1635–1642.
- [TB96] TURK G., BANKS D.: Image-guided streamline placement. In *ACM SIGGRAPH 96 Conference Proceedings* (1996), pp. 453–460.
- [Tuf83] TUFTE E. R.: *The Visual Display of Quantitative Information*. Graphics Press, Cheshire, CT, 1983.
- [TvW99] TELEA A., VAN WIJK J.: Simplified representation of vector fields. In *Proceedings IEEE Visualization '99* (1999), pp. 35–42.
- [vW02] VAN WIJK J. J.: Image based flow visualization. *ACM Transactions on Graphics* 21, 3 (2002), 745–754.
- [vW05] VAN WIJK J. J.: The value of visualization. In *Proceedings of the IEEE Visualization '05* (2005), IEEE Computer Society, pp. 79–86.
- [ZYZL09] ZHANG E., YEH H., ZHONGZANG L., LARAMÉE R. S.: Asymmetric tensor analysis for flow visualization. *IEEE Transactions on Visualization and Computer Graphics* 15, 1 (2009), 106–122.